Virag Mishra

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Profile
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Education

Ph.D. 2020
Department of Mechanical Engg. **Victoria**
University of Victoria **Canada**

M.S. (by research) 2009
Department of Ocean Engg., Indian Institute of Technology **(IIT)** Madras

Bachelor of Engineering. (Mechanical) 2005
Department of Mechanical Engg. Raipur
Pt. Ravi Shankar Shukla Univ. India

Ph.D. | **Data-driven hydrodynamic models for Heaving Wave Energy Converters**

- CFD simulations were used to generate high-fidelity datasets for a data-driven modelling framework of two-body WECs, using openFoam.

M.S. | **CFD analysis of underwater vehicles**

- Developed a CFD methodology in ANSYS Fluent to compute longitudinal and translational added mass of five axisymmetric underwater vehicles. Implemented and customized non-linear k-ε, modified k-ε and modified k-ω turbulence models using UDFs for improved hydrodynamic prediction.

Employment

Research Associate March 2025 -
Oregon State University Corvallis, **USA**

Hydrodynamic Engineer July 2020 - Aug 2024
Carnegie Clean Energy Ltd. Perth, **Australia**

CAE Analyst Aug 2011 - March 2013
TATA Technologies Ltd. Pune, India

CFD Engineer July 2009 - May 2011
ESI-Group Pune, India

Current Research Summary

• **Hydrodynamic modelling and tank testing of Wave Energy Tech.** 2025-2026

Funded By:	**U.S. Navy**
Commercial Partner:	**Dolphin Labs**
Objective:	Development of a numerical and scaled model for sub-surface wave energy conversion systems

• **Hydrodynamic analysis of hybrid OWC-floating wind turbine platform** 2025-2026
Academic Partners: **University of Victoria, University of the Basque Country**
Objective: Vibration mitigation of Floating Offshore Wind Turbine (FOWT)

Previous Project Summary

Funded By: **Carnegie Clean Energy Ltd**
Project value: $22.5 million (EuropeWave Phase 1-3)
Years: 2020-2024

- **Development of numerical models**
- **Implementation of control algorithms**
- **Hull shape optimization**
- **Variable depth operation**
- **Integrated CFD & LTD workflow**
- **Met-Ocean & Weather Window**
- **Tank & Open water testing**

Previous Grants & Scholarships

NAVSEA-Energy Harvester R&D Program 2025-
EuropeWave PCP program 2022-2024
TEAMER-RFTS 12 Technical Support funding 2024
Natural Sciences & Engineering Research Council of Canada **(Doctoral)** 2013-2017
Natural Sciences & Engineering Research Council of Canada **(Engage grant)** 2018
Mitacs **(Accelerate program)** 2019
Ministry of Human Resource Development **(MHRD)**, Govt. of India 2006-2009

Mentorship (Oregon State University)

Team	**Involvement**	**Duration**
Human powered Submarine Team	Design of submarine and propulsion systems	Nov 2025-
Capstone Project	Design and testing of scaled Two-Body submerged WEC	Sept 2025-March 2026
Capstone Project	Design and prototyping of rotory PTO for WEC	Jan 2026-
Capstone Project (E-Campus)	CAD Design and material selection of submergedWEC	Jan 2026-

Last updated: October 15, 2026